JAVA ENTERPRISE PRACTICALS

Practical 2: Login Page Using Servlet

Environment: Apache NetBeans IDE · GlassFish Server · JDK 21

1. Aim

To create a login page using HTML and Java Servlet. If the username and password are correct, display the message “Hello”; otherwise, display “Login Failed”.

2. Software Requirements

- Java JDK 21
- Apache NetBeans IDE
- GlassFish Server
- Web browser
- HTML and Java

3. Theory

A Servlet is a Java server-side component that processes client requests and generates responses. In this practical, an HTML login form accepts a username and password. The form sends the values to a Java Servlet using HTTP POST. The Servlet checks the credentials and displays the appropriate message.

For this classroom example, the valid credentials are:

- **Username:** admin
- **Password:** 1234

These are demonstration credentials only, not suitable for a production login system.

4. Algorithm

1. Create an HTML login form.
2. Accept the username and password from the user.
3. Submit the form to the Servlet using HTTP POST.
4. Retrieve the values using `request.getParameter()`.
5. Compare the entered credentials with the correct credentials.
6. If both credentials match, display “Hello”.
7. Otherwise, display “Login Failed”.
8. End the program.

5. Create the Project in NetBeans

1. Open Apache NetBeans.
2. Select **File** → **New Project**.
3. Select **Java Web** → **Web Application**.
4. Enter the project name: **LoginServlet**.
5. Select GlassFish Server.
6. Choose the Java EE/Jakarta EE version supported by the installed server.
7. Click **Finish**.

If you already have a project for this practical, you can use it instead.

6. Project Structure

The project should contain:

- **Web Pages**
  - `index.html`
- **Source Packages**
  - `com.login`
    - `LoginServlet.java`

7. Program 1: index.html

Under Web Pages, create an HTML file named `index.html`. Enter the following code:

```
<!DOCTYPE html>
<html>
<head>
    <title>Login Page</title>
</head>
<body>
    <h2>Login Page Using Servlet</h2>

    <form action="login" method="post">

        <label>Username:</label>
        <input type="text" name="username" required>
        <br><br>

        <label>Password:</label>
        <input type="password" name="password" required>
        <br><br>

        <input type="submit" value="Login">

    </form>
</body>
</html>
```

Explanation

- `form` creates the login form.
- `action="login"` sends the form to the Servlet mapped to /login.
- `method="post"` submits the form using HTTP POST.
- `name="username"` identifies the username field.
- `name="password"` identifies the password field.
- `type="password"` masks the password characters on screen.
- `required` prevents empty fields from being submitted through the browser form.

8. Program 2: LoginServlet.java

Create a Java package named `com.login` under Source Packages. Inside that package, create a Servlet named `LoginServlet`.

Use the following code for GlassFish 6 or later:

```
package com.login;

import java.io.IOException;
import java.io.PrintWriter;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
```

```
@WebServlet("/login")
public class LoginServlet extends HttpServlet {

    @Override
    protected void doPost(HttpServletRequest request,
                          HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html;charset=UTF-8");

        PrintWriter out = response.getWriter();

        String username = request.getParameter("username");
        String password = request.getParameter("password");

        if ("admin".equals(username)
                && "1234".equals(password)) {

            out.println("<html><body>");
            out.println("<h2>Hello</h2>");
            out.println("</body></html>");

        } else {

            out.println("<html><body>");
            out.println("<h2>Login Failed</h2>");
            out.println("</body></html>");
        }
    }
}
```

Compatibility note: GlassFish 5 uses `javax.servlet` imports. GlassFish 6 and later use `jakarta.servlet` imports. Use the namespace supported by your configured server.

9. Important Code Explanation

Code	Purpose
@WebServlet("/login")	Maps the Servlet to the /login URL.
extends HttpServlet	Makes the class an HTTP Servlet.
doPost()	Processes HTTP POST requests.
request.getParameter()	Retrieves the submitted username and password.
if-else	Checks credentials and selects the response.
&&	Requires both conditions to be true.
response.setContentType()	Sets the response type to HTML.
response.getWriter()	Obtains a writer to send output to the browser.

10. How the Login Validation Works

The Servlet retrieves the values:

```
String username = request.getParameter("username");
String password = request.getParameter("password");
```

It checks both credentials:

```
if ("admin".equals(username)
        && "1234".equals(password))
```

Both comparisons must be true for the program to display “Hello”. Using a constant string’s `equals()` method also avoids a `NullPointerException` if a parameter is missing.

11. Execution Steps

1. Save both files using **Ctrl + S**.
2. Right-click the project and select **Clean and Build**.
3. Check the Output window for errors.
4. Right-click the project and select **Run**.
5. Wait for GlassFish to deploy the application.
6. Open the `index.html` page in the browser if it does not open automatically.
7. Enter the username **admin**.
8. Enter the password **1234**.
9. Click **Login**.
10. Confirm that “Hello” appears.
11. Test incorrect credentials and confirm that “Login Failed” appears.

The application URL commonly resembles:
`http://localhost:8080/LoginServlet/`
The port and context path may vary depending on the GlassFish configuration.

12. Expected Output

Case 1: Correct Credentials

Input:

```
Username: admin
Password: 1234
```

Output:

```
Hello
```

Case 2: Incorrect Credentials

Input:

```
Username: student
Password: 1111
```

Output:

```
Login Failed
```

Case 3: Correct Username but Incorrect Password

Input:

```
Username: admin
Password: 1111
```

Output:

```
Login Failed
```

13. Common Errors and Solutions

Error: package jakarta.servlet does not exist

Check the GlassFish version and project server configuration. Ensure that the correct Servlet API is available through the server runtime.

Error: HTTP 404

Check the form action, Servlet URL mapping, and deployed application URL.

Error: GlassFish does not start

Verify the Java platform configured for the server and check the NetBeans server logs.

Error: Build failed

Read the first compiler error in the Output window and correct the indicated problem.

14. Viva Questions and Answers

Q1. What is a Servlet?

A Servlet is a Java server-side component that handles client requests and generates responses.

Q2. What is the purpose of doPost()?

It handles HTTP POST requests submitted by a client.

Q3. What does request.getParameter() do?

It retrieves a value submitted through a form field.

Q4. Why is the password field type set to password?

It masks the characters displayed on screen. It does not encrypt the password.

Q5. What does && mean in Java?

It is the logical AND operator. Both conditions must be true.

Q6. What is the purpose of @WebServlet("/login")?

It maps the Servlet to the /login URL pattern.

Q7. What is the difference between authentication and authorization?

Authentication verifies a user's identity. Authorization determines what the user is allowed to access.

Q8. Is hardcoding a password secure?

No. Hardcoded credentials are used here only for demonstration. Real applications should use secure password hashing and proper authentication mechanisms.

Q9. What happens if only the username is correct?

The full condition evaluates to false, and the Servlet displays "Login Failed".

Q10. What is the role of GlassFish?

GlassFish runs the Java web application and manages the Servlet container.

15. Result

The login page was developed using HTML and Java Servlet. When the correct username and password are entered, the application displays "Hello"; otherwise, it displays "Login Failed".

16. Quick Revision

Remember these five key points:

1. `action="login"` sends the form to the Servlet.
2. `@WebServlet("/login")` defines the Servlet mapping.
3. `request.getParameter()` retrieves the submitted credentials.
4. `&&` checks that both credentials match.
5. The Servlet displays "Hello" for valid credentials and "Login Failed" otherwise.